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V: < 16p
3: 16-22p
4: 23-30p
5: > 30p

Biometrics Recognition DI 2009

Exam 27 Oct. 2016

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①

Choose any three & explain:

- a) 3p
- Universality
 - Uniqueness
 - Permanence/stability
 - Collectability
 - Performance
 - Acceptability
 - Vulnerability

b)

2p

	univ.	unig.	stab.	collect	Perf.	accept.	vulner.
Fingerprint	H	H	H	L	H	M	M/L
Face	H	M	L	H	M/L	H	H
Voice	H	M/L	L	M/H	M/L	H	H

Other answers accepted if reasoned appropriately

- c) 2p
- Universality: { affect Failure to Acquire (FTA) / Failure to Enroll (FTE)
 - Collectability

- Uniqueness: { affect FAR/FRR
- Performance:

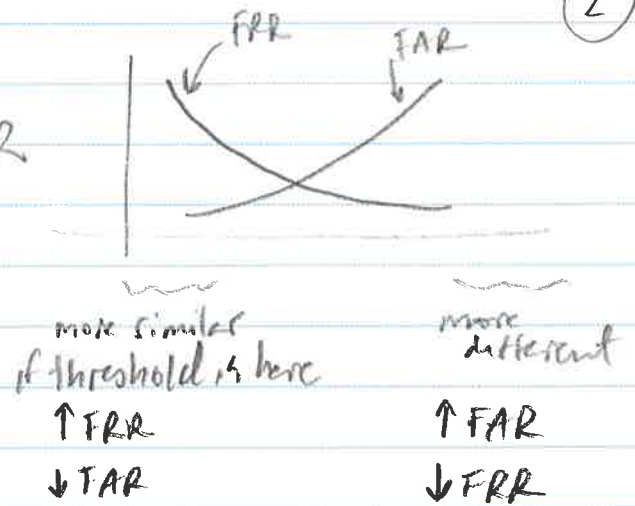
- Stability: increases FRR of a user over time

- Acceptability: affect "success" of the application



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② a) Red: FRR
2p Blue: FAR



b) High convenience: low FRR → choose higher T

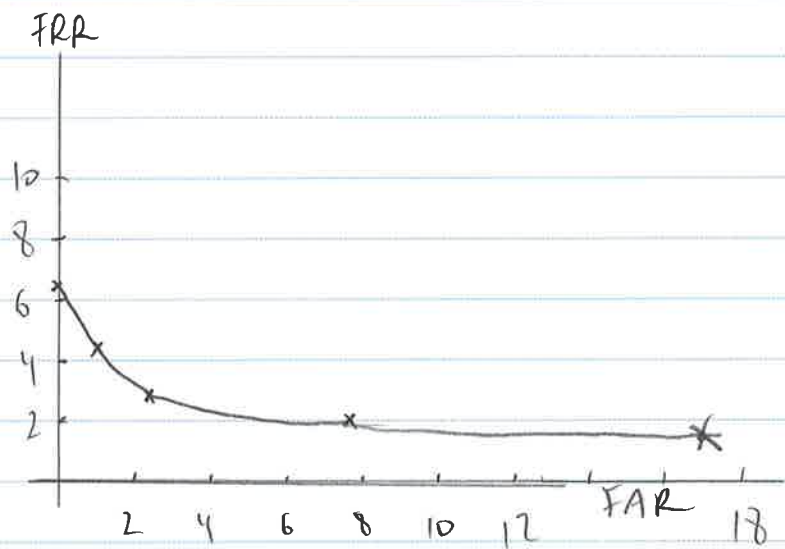
2p To avoid FAR > 10%, $T < 9.3 / 9.4$

At this T, FRR is ~ 2%

c)

2p

T	FRR (red)	FAR (blue)
4	16	0
5	10.5	0
6	6.5	0
7	4.5	1
8	3	2.5
9	2	7.5
10	1.5	17





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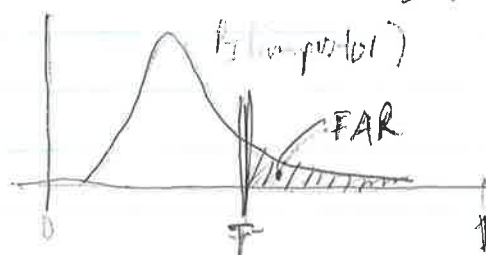
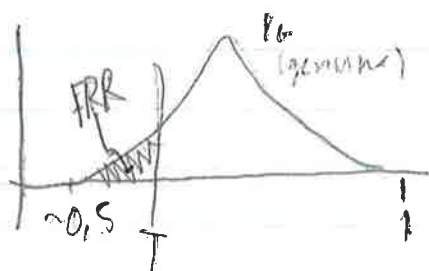
③ von fraction mode, 1200 genuine matchings
1800 impostor

a) Similarity measure, $S > T$ accept
 $S \leq T$ reject

9p.

$$FRR(T) = \sum_{S=0}^T P_G(S)$$

$$FAR(T) = \sum_{S=T}^1 P_I(S)$$

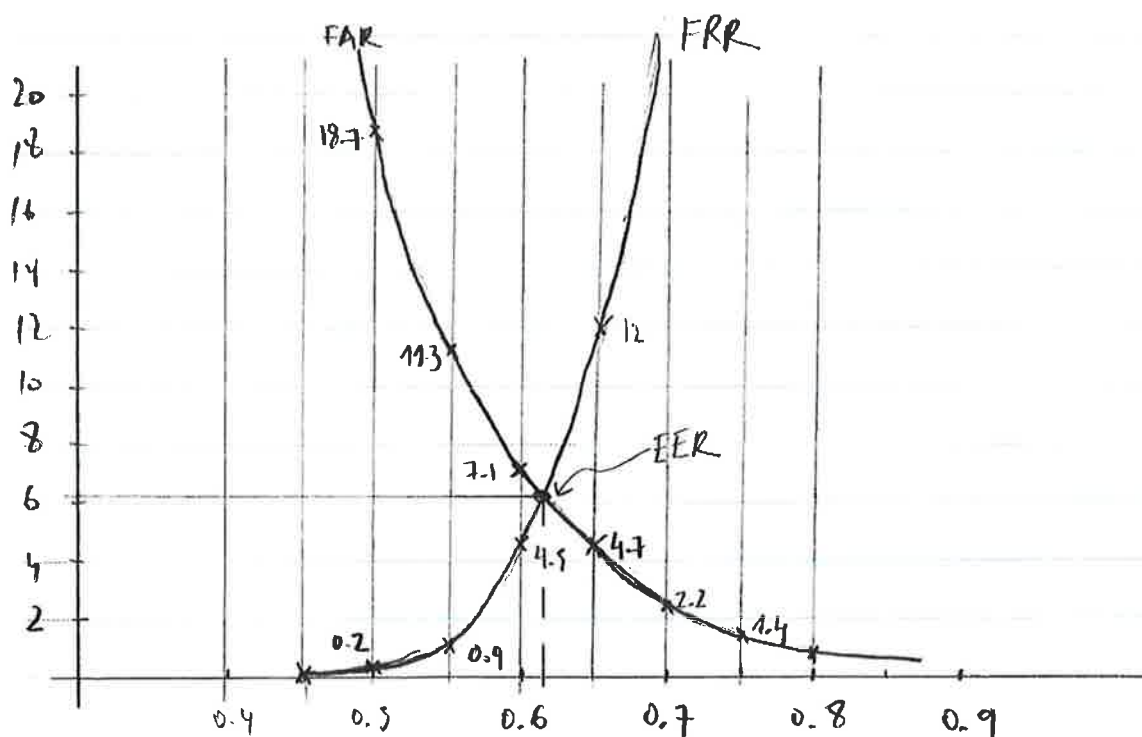


- overlap is from T to 0.5

T	genuine (amount < T)	impostor (amount > T)
0.45	0	448
0.5	2	337
0.55	11	203
0.6	54	127
0.65	144	85
0.7	337	40
0.75	607	26
0.8	895	16

=>

FRR (% below T)	FAR (% above T)	T
0/1200 = 0, = 0%	448/1800 = 0.249 = 24.9%	0.45
2/1200 = 0.002 = 0.2%	337/1800 = 0.187 = 18.7%	0.5
0.9%	11.3%	0.55
4.5%	7.1%	0.6
12%	4.7%	0.65
29.1%	2.2%	0.7
50.6%	1.4%	0.75
74.6%	0.9%	0.8





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(4)

b) From figure: EER $\sim 6 / 6.1\%$

1P

$$T \approx 0.61 / 0.62$$

c) max. 1% FAR when FRR = 10%

2P

$$FRR = 10\% \text{ is at } T = 0.64$$

At this point FAR = 4.8% > 1%

cannot meet the requirement

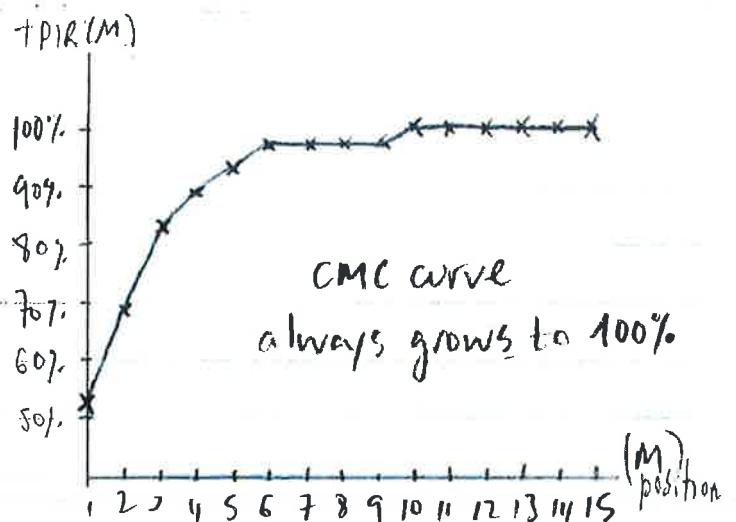
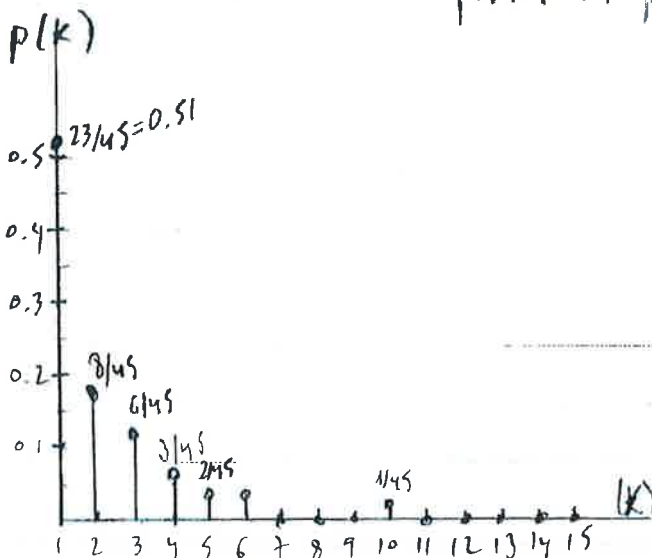
(4) Identification mode, 15 persons, $15 \times 3 = 45$ tests in total

a) 4P

Position	1	2	3	4	5	6	7	8	9	10	11	...	15
# of correct ID	23	8	6	3	2	2	0	0	0	1	0	...	0
$P(K)$	$23/45$	$8/45$	$6/45$	$3/45$	$2/45$	$2/45$	0	0	0	$1/45$	0	...	0
TPIR(M)	$\frac{23}{45}$ 0.51 51%	$\frac{23+8}{45}$ 0.69 69%	$\frac{23+8+6}{45}$ 0.822 82.2%	...	...	...	...	...	...	...	...	...	...
	51%	69%	82.2%	89%	93%	98%	98%	98%	98%	100%	100%	...	100%

$P(K)$ = probability that the correct ID is in position K of the output list

$TPIR(M) = \sum_{K=1}^M P(K)$ = probability that the correct ID is in the first M positions of the output list



(5)



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b) Requirement $TPIR(M=3) > 80\%$

2p.

We have $TPIR(3) = 82.2\% > 80\%$.

so the system meet the requirement

⑤ a) x Highest frequency: cosine 3 (more oscillations in the same number of samples)

3p.

x least/smallest frequency: cosine 2

x Highest amplitude: cosine 3 (goes between $+1.5/-1.5$)

x Smallest amplitude: cosine 1 and 2

(both go between $+1/-1$ in the y-axis)

b) Gaussian filter: smooths the image/eliminate sharp edges

3p.

Higher size: Filter 2 (smooths more than filter 1)

c) Poor quality: top part because

2p.

- ridges are more blurry / are less sharp

- lost of "false" M points detected before enhancing the image

Enhancement

- successful because improves "smoothness" of the ridges and make them more "continuous" in bad quality regions

- lost of "false" M points disappear after enhancement

Detection is never 100% perfect anyway! You always miss some "real" M points, and detect some "false" ones